

|                                    |                                               | L1<br>known rule | L2<br>seen worlds | L3<br>unseen worlds | L4<br>with prior | params |
|------------------------------------|-----------------------------------------------|------------------|-------------------|---------------------|------------------|--------|
| <b>Standard models</b>             |                                               |                  |                   |                     |                  |        |
| <b>a</b>                           | CNN                                           | 100              | 22.6              | 18.9                | 10.0             | 116k   |
| <b>c</b>                           | transformer, 2 layers (d64)                   | 100              | 28.7              | 25.8                | 5.3              | 166k   |
| <b>c</b>                           | transformer, 4 layers (d128)                  | 100              | 86.7              | 85.6                | 78.6             | 925k   |
| <b>c</b>                           | transformer, 6 layers (d192)                  | 100              | 39.5              | 37.3                | 43.1             | 2.87M  |
| <b>c</b>                           | transformer, 8 layers (d256)                  | 100              | 54.1              | 52.7                | 0.2              | 6.58M  |
| <b>g</b>                           | diffusion                                     | 24.9             | 0.0               | 0.0                 | 0.0              | 86k    |
| <b>Spatial locality (Sec. 3)</b>   |                                               |                  |                   |                     |                  |        |
|                                    | transformer, raster RoPE, 2 layers (d64)      | 39.1             | 20.3              | 17.2                | 1.3              | 100k   |
|                                    | transformer, axial RoPE, 2 layers (d64)       | 100              | 19.2              | 15.8                | 0.9              | 100k   |
| <b>Temporal locality (Sec. 4)</b>  |                                               |                  |                   |                     |                  |        |
|                                    | transformer, 2 layers (d64) + last-frame 3×3  | 100              | 99.9              | 99.9                | 73.2             | 167k   |
|                                    | transformer, 4 layers (d128) + last-frame 3×3 | 100              | 99.9              | 99.9                | 97.2             | 926k   |
| <b>d</b>                           | 2D-conv transformer, 2 layers                 | 100              | 89.1              | 88.8                | 79.9             | 142k   |
| <b>e</b>                           | KV-shift transformer, 1 memory read           | 100              | 100               | 100                 | 42.2             | 2.6k   |
| <b>e</b>                           | KV-shift transformer, 4 memory reads*         | 100              | 100               | 100                 | 99.5             | 9.8k   |
| <b>f</b>                           | 3D-conv transformer, 1 memory read            | 100              | 100               | 100                 | 71.5             | 3.8k   |
| <b>f</b>                           | 3D-conv transformer, 4 memory reads           | 100              | 100               | 100                 | 99.5             | 11k    |
| <b>b</b>                           | 3D-conv CNN                                   | 100              | 99.8              | 99.7                | 98.8             | 18k    |
| <b>Temporal stability (Sec. 5)</b> |                                               |                  |                   |                     |                  |        |
|                                    | per-frame noise + backward encoder + anchor   | 32.0             | 6.1               | 4.5                 | 0.1              | 30k    |
| <b>h</b>                           | same denoiser, frame-ordered sampling         | 100              | 30.9              | 27.9                | 24.8             | 30k    |
| <b>h</b>                           | same denoiser, order-matched training         | 100              | 99.3              | 99.4                | 93.1             | 30k    |